Lab 1

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Introduction

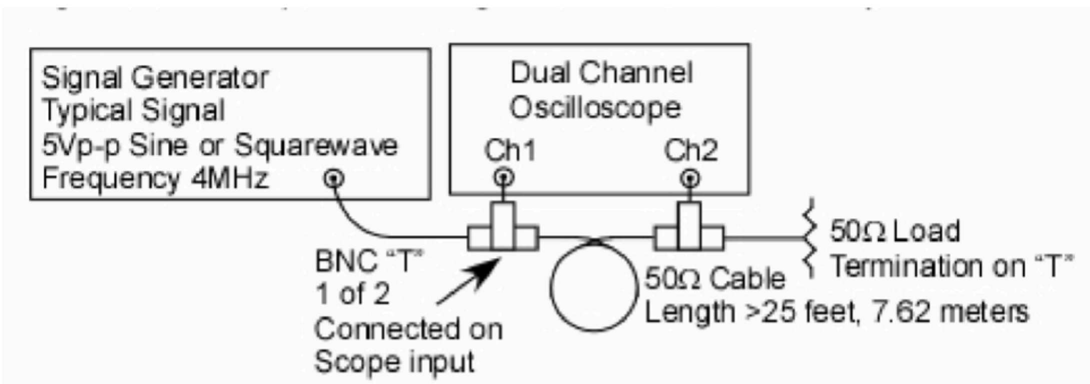
The time delay (t_d) is the amount of time it takes for the signal to travel from the input of the cable to the output. If the physical length of the cable is known, the velocity of propagation (v_p) can be calculated as $v_p = \frac{l}{t_d} = \frac{length\ of\ the\ cable}{measured\ time\ delay}$. For coaxial cables the signal travels at about 66% of the speed of light ($v_p \approx\ 2x10^8 \frac{m}{s}$)- meaning that a 25 foot cable should produce a delay of around $2x10^8 \frac{m}{s}$.

For sinusoidal signals, time delay can be also described as a phase shift. Phase shift tells us how much the output wave is shifted in angle relative to the input wave, and depends on both time delay and frequency. Phase shift is given by $phase\ shift\ (degrees)\ =\ 360^o * t_d * f)$, where *f* is the signal frequency. It can be seen that as frequency increases the phase shift increases, which is why transmission line effects have a greater impact at higher frequencies.

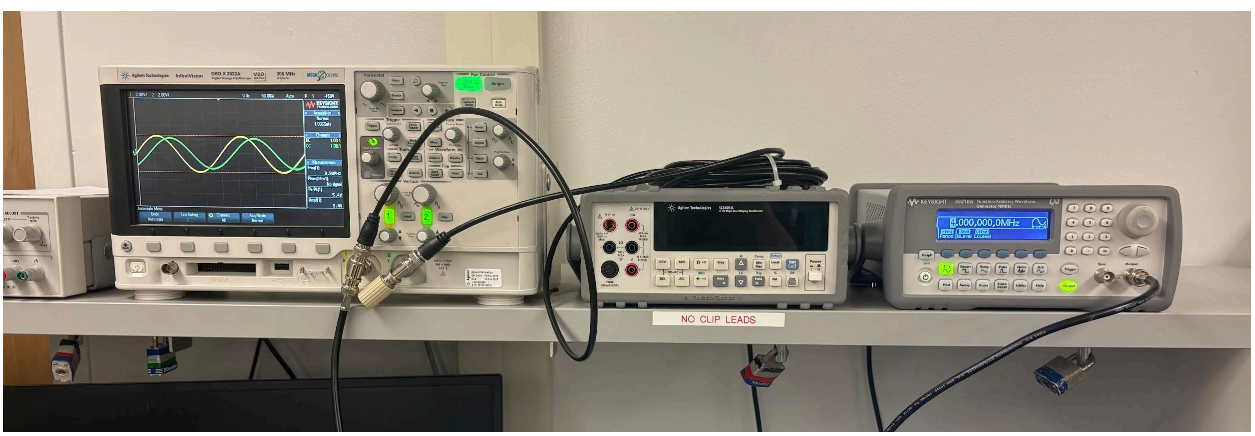
Additionally, the concept of wavelength (the physical distance a signal travels during one full cycle) is introduced in this lab. Wavelength (λ) is related to velocity and frequency by $\lambda = \frac{v_p}{f}$ and it can be seen that as frequency increases, wavelength decreases.

In this lab we studied how an electrical signal changes as it travels through a 50-ohm coaxial cable that is 25 feet long. When a signal travels through a cable, it does not arrive instantly at the other end. Instead it experiences a time delay, a phase shift, and sometimes a small change in voltage. These effects become more noticeable as the signal frequency increases, especially above 1 MHz. Therefore knowing how the specific transmission medium (coaxial cables, broadcast signals, etc.) impacts the transmitted signal is important (especially as the signal frequency increases to above 1MHz).

1. Lab schematic and physical setup: In the setup a length of 50-ohm coax cable (length less than 5 or 6 feet), a length of 50-ohm (50) coax cable (25 feet and referred to below as the transmission line or T-line), two Coax BNC-T connectors, and one 50 coaxial load. Note: Some modern oscilloscopes have the option to switch their input impedances to 50ohms for high frequency work, in such cases Ch2 (NOT Ch1) can be set for 50 AND the BNC T and 50ohm load termination removed but this is your choice but do have two 50 in parallel.



IMG 1. *Diagram for test set up for measuring transmission line time delay*

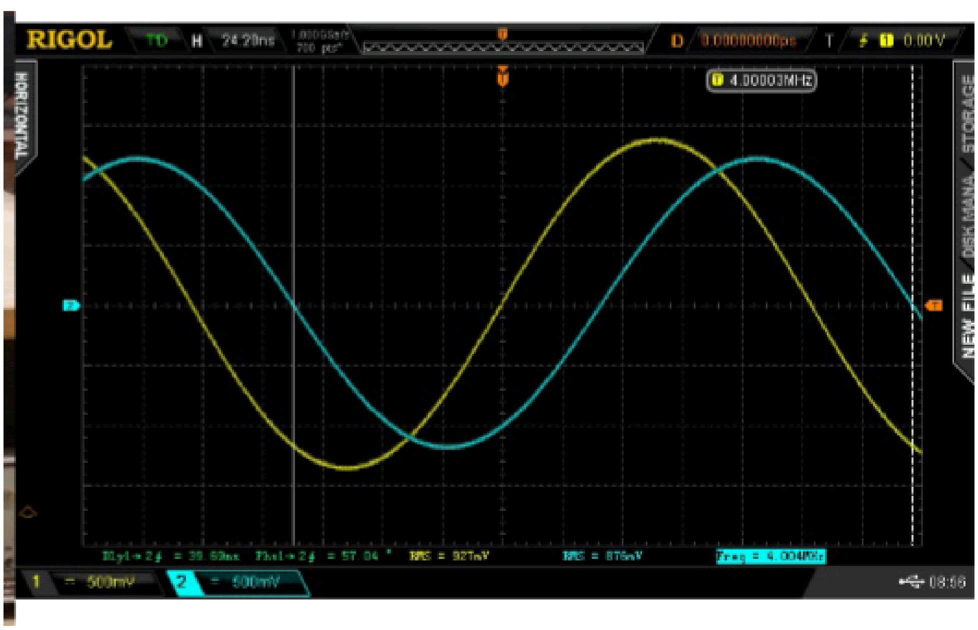


IMG 2. *Our set up.*

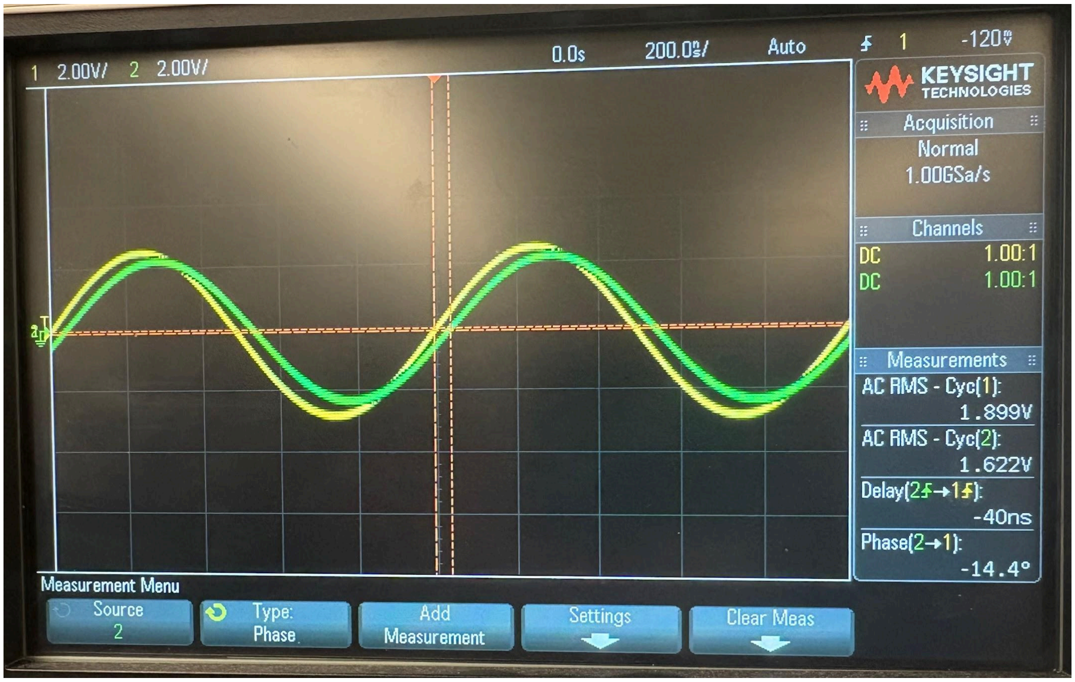
2.Data

Frequency (MHz)	Time Delay (ns)	Phase Shift (degrees)	Vin Channel 1 (Vrms)	Vout Channel 2 (Vrms)
1	40	14.4	1.899	1.622
2	40	29	1.902	1.598
3	40	43	1.885	1.619
4	39	56	1.884	1.619
5	38	69	1.813	1.527
6	39	85	1.792	1.527
7	39	95	1.826	1.563
8	39	115	1.814	1.550
9	39	126	1.817	1.520
10	39	140	1.827	1.538

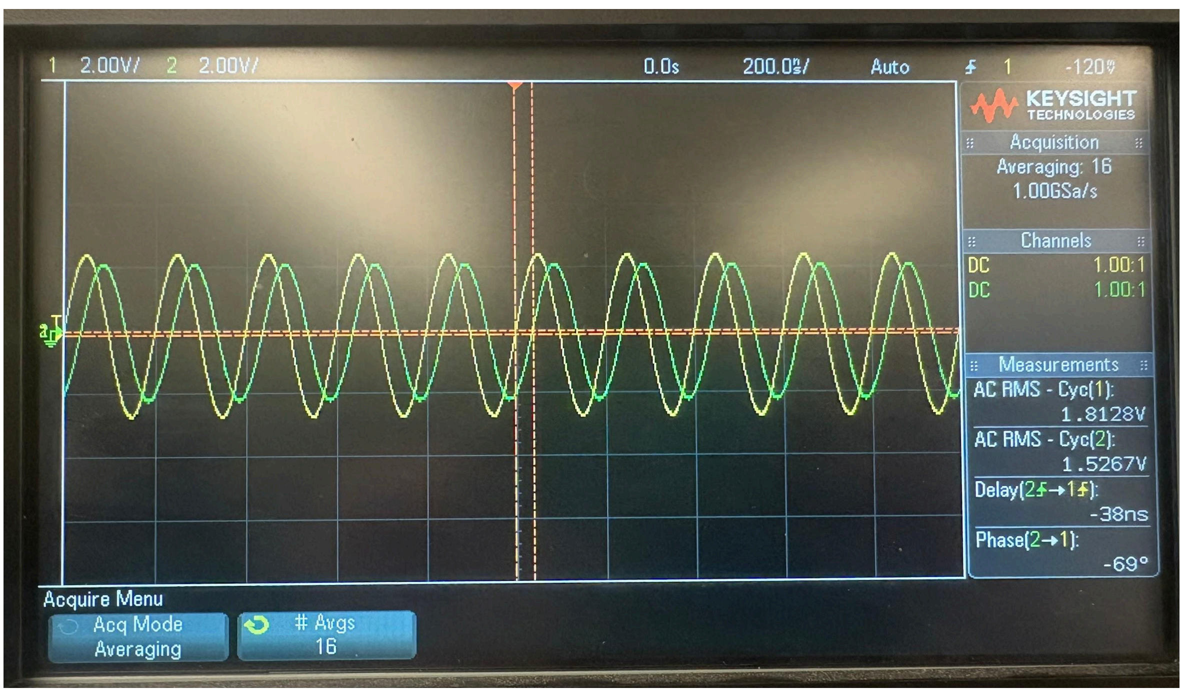
Table 1. *Time delay and phase shift measurements with an input of a sine wave. Note that time cannot be negative. To adjust for the resulting negative phase shift, 360 degrees can be added to each phase shift data point. The generator used was limited to 10Mhz, hence the frequency range of the data. Pictures were taken for 1Mhz,5Mhz, and 10Mhz.*



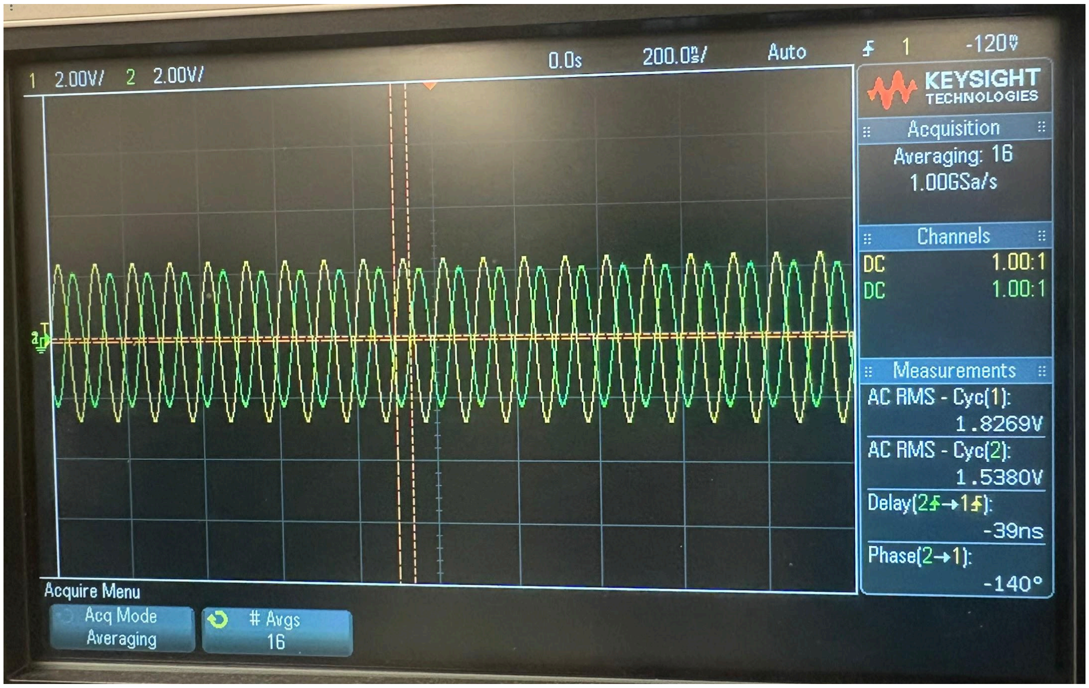
IMG 3. *Predicted results*



IMG 4. *1Mhz*



IMG 5. *5MHz*



IMG 6. *10Mhz*

Results

By calculation the velocity of propagation for the electronic signal in the coaxial cable which should be about 0.66c or 2/3 the speed of light or ~200 million meters/sec, (2x108 m/s). For a 25 foot, 7.62 meter, cable length this will be around 38 nanoseconds which your data should confirm based upon the cable type and its Velocity of Propagation, i.e. RG-58 is about 2/3 or 66% of the Speed of Light or (2/3) x (3 x108 meters/sec) or 200 x106 meters/sec.

Key formulas:

- $v_p = \frac{l}{t_d} = \frac{length\ of\ the\ cable}{measured\ time\ delay}$.
- $phase\ shift\ (degrees)\ =\ 360^o * t_d * f$
- $\lambda = \frac{v_p}{f}$

Note that for coaxial cables the signal travels at about 66% of the speed of light ($v_p \approx 2x10^8 \frac{m}{s}$).

A. Confirm the physical cable length

Velocity of Propagation confirmation for 5MHz:

$v_p = \frac{l}{t_d} = \frac{length\ of\ the\ cable}{measured\ time\ delay}$ is re-arranged to $l = v_p \times t_d$

$7.62m = v_p \times (38ns)$

$v_p = 2 * 10^8\ m/s$

Confirm the physical cable length:

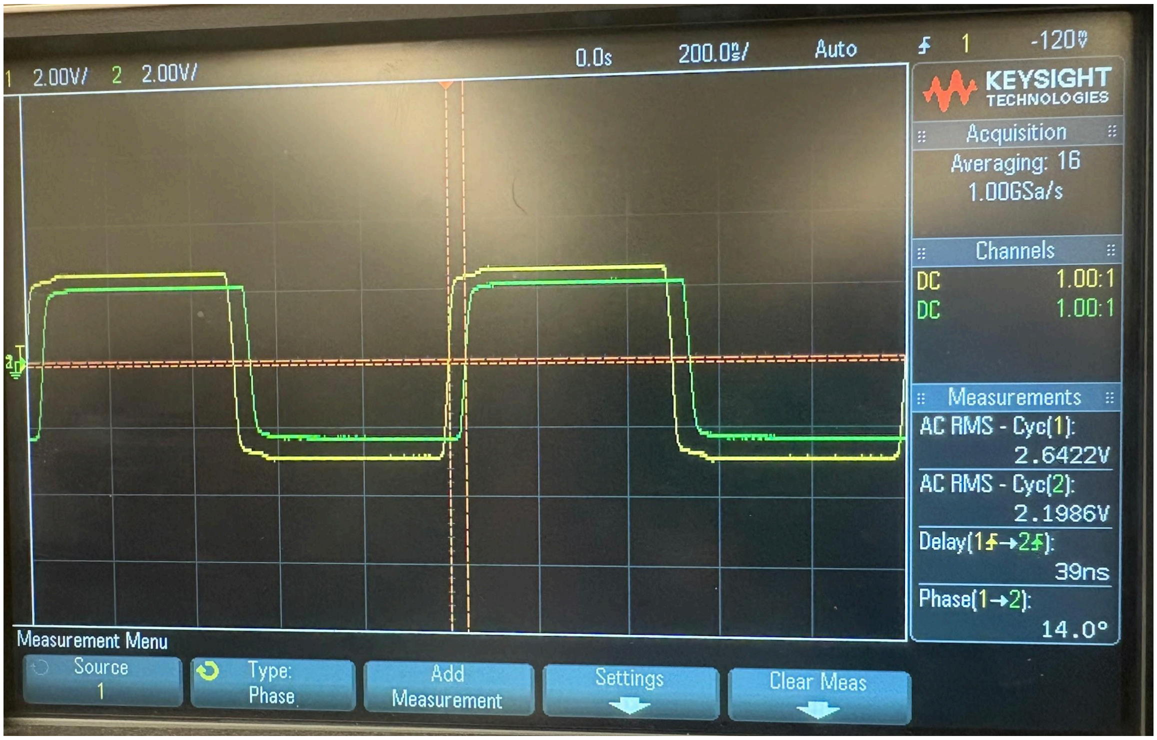
$l = v_p \times t_d$ =2 $* \ 10^8\ m/s$ (39x10^-9)= 8 meters

3.Data

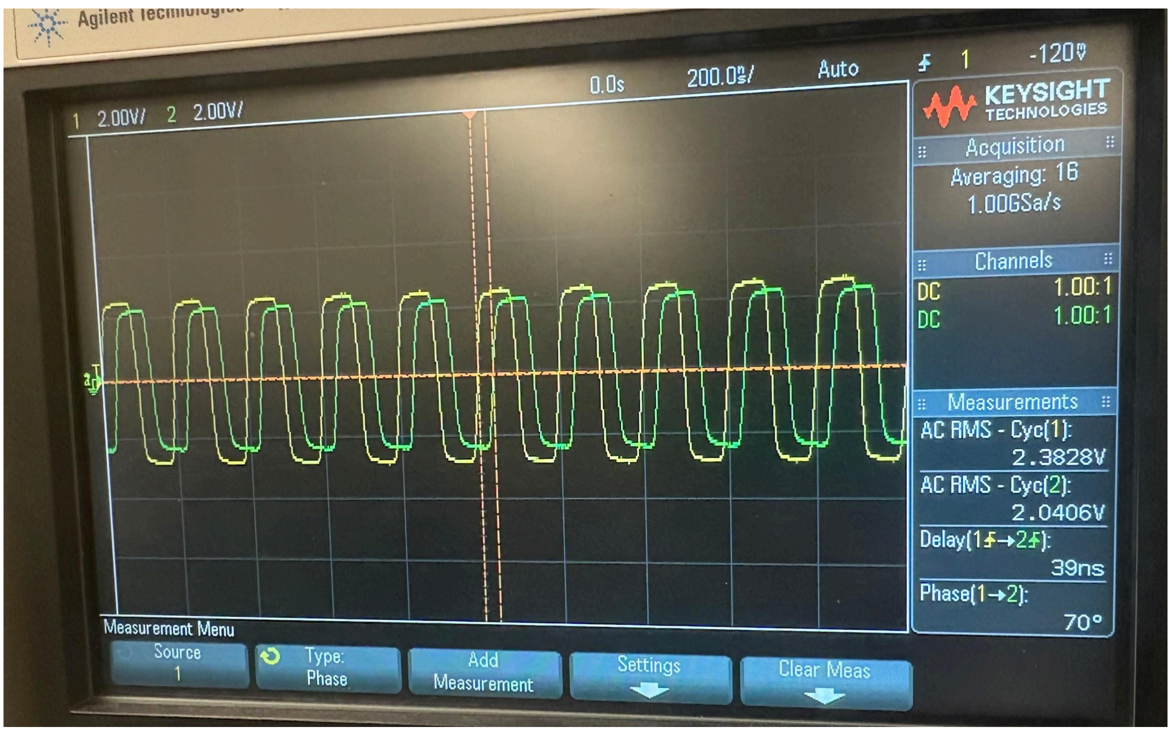
Change the signal to a square wave and record the time delay (we usually do not measure phase for digital signals) for three representative frequencies (i.e. 4, 8 and 12 MHz or 5, 10 and 15 MHz) as a visual reminder of the laboratory measurement data. What do you notice about the data?

Frequency (MHz)	Time Delay (ns)	Vin Channel 1 (Vrms)	Vout Channel 2 (Vrms)
1	39	2.642	2.199
5	39	2.383	2.041
10	39	2.038	1.752

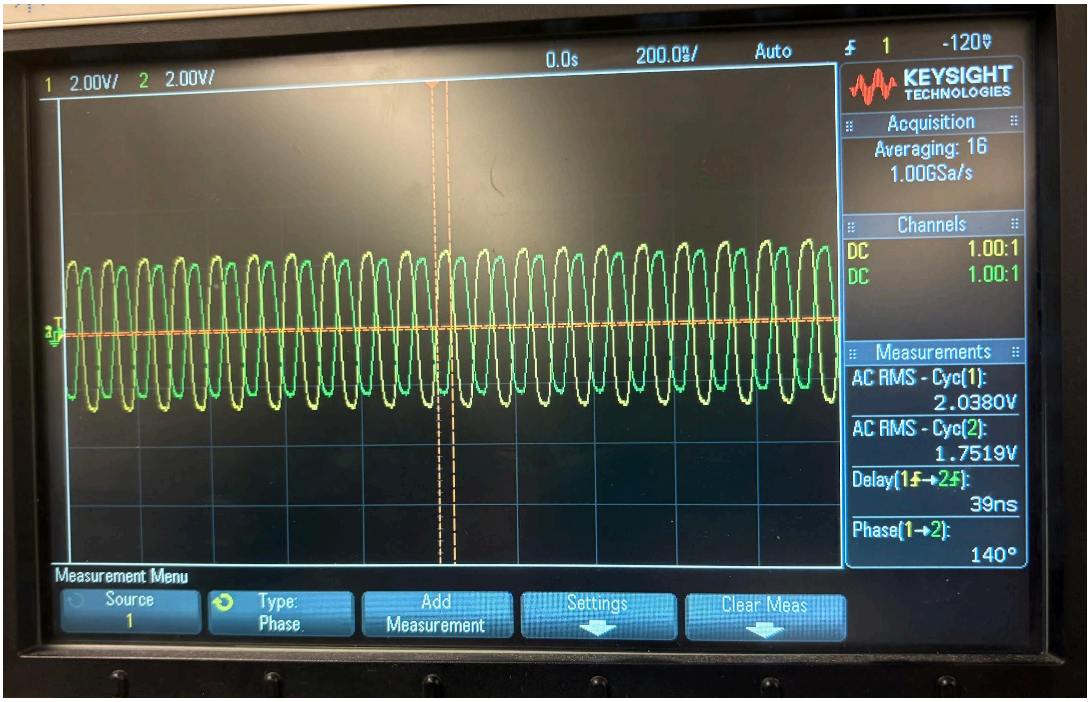
Table 2. *Time delay and phase shift measurements when the input is a square wave*



IMG 7. 1 Mhz



IMG 8. 5Mhz



IMG 9. 10Mhz

Calculate the wavelength of lambda and 25-foot cable length in lambda. Use vp = 0.66c or (2/3)c at the speed of light or ~200 million meters/sec, (2x108 m/s).

Determining Lambda:

$$\lambda = \frac{v_p}{f}$$

Note that for coaxial cables the signal travels at about 66% of the speed of light ($v_p \approx 2x10^8 \frac{m}{s}$*).*

$$\lambda = \frac{2*10^8 m/s}{1*10^6 1/s} = 2 * 10^2 = 200m$$

Frequency (MHz)	Lambda from frequency (m)	25ft Cable Length (in lambda)
1	200.000	0.125

2	100.000	0.250
3	66.667	0.375
4	50.000	0.500
5	40.000	0.625
6	33.333	0.750
7	28.571	0.875
8	25.000	1.000
9	22.222	1.125
10	20.000	1.250

Table 3. *Wavelength calculations*

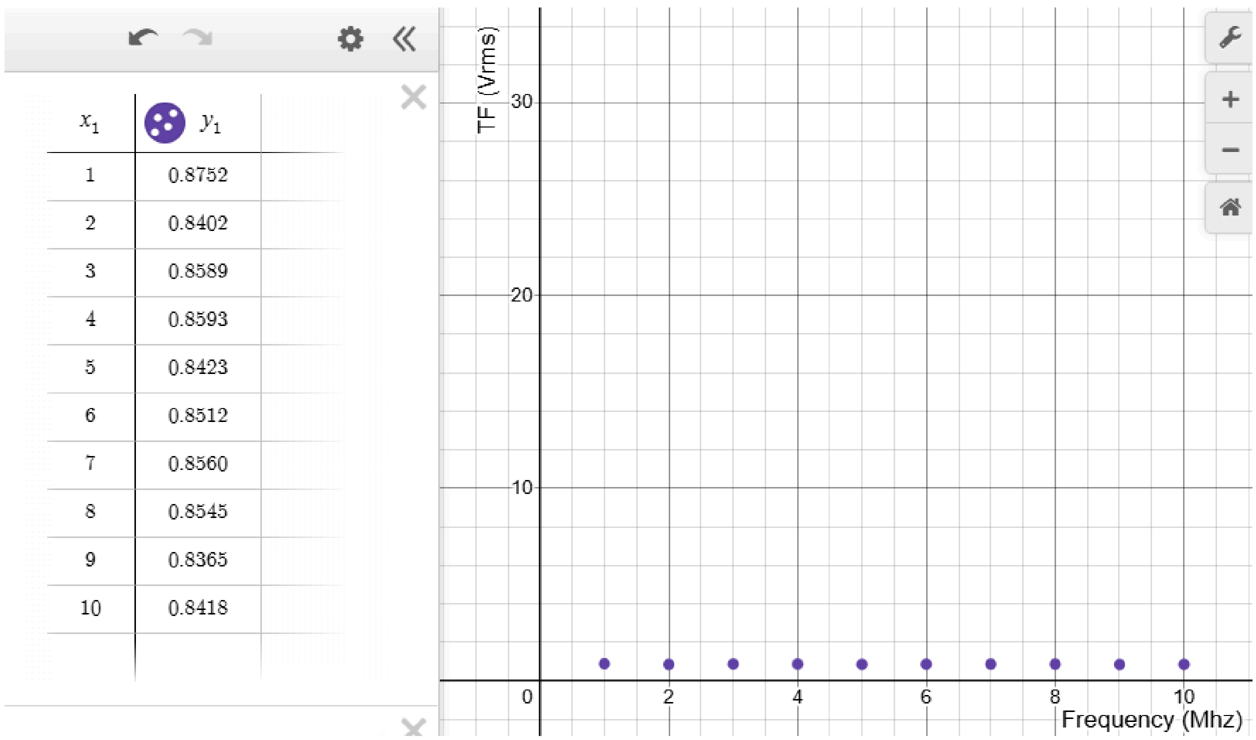
4. Plotting the data

For the sine wave case (part 2 data), calculate and plot the Vout/Vin, transfer function of the cable vs frequency. What do you notice about the data? Does it make sense based upon your previous lab work?

Frequency (MHz)	Vin Channel 1 (Vrms)	Vout Channel 2 (Vrms)	$\frac{V_{out}}{V_{in}}$
1	1.899	1.622	0.8752
2	1.902	1.598	0.8402
3	1.885	1.619	0.8589
4	1.884	1.619	0.8593

5	1.813	1.527	0.8423
6	1.792	1.527	0.8512
7	1.826	1.563	0.8560
8	1.814	1.550	0.8545
9	1.817	1.520	0.8365
10	1.827	1.538	0.8418

Table 4. *Transfer function*



IMG 10. *Graph of transfer function (*$\frac{V_{out}}{V_{in}}$ *in Vrms) vs Frequency (in MHz).*

When we plot the transfer function verses the frequency, the graph stayed almost flat. This means that the output voltage stayed nearly the same as the input voltage over the range of the tested frequencies. Basically, the cable did not significantly reduce the signal strength. At higher frequencies the TF ratio might decrease slightly, meaning that the signal could lose a small amount of amplitude as frequency increases. This small loss was actually not observed in our data, but might be observed in data with higher frequencies. The voltage did not change much, showing us that the main effect of the coaxial cable is not reducing the signal size but instead delaying the signal in time and shifting its phase.

Conclusion

In this lab, we were able to see how the permittivity of a transmission medium can affect a signal. In addition, we were able to calculate inherent properties about the transmitted signal using the information we recorded. Knowing how to capture and calculate values from these signals is especially important when designing a communication system in order to compensate for issues unique to high frequency transmissions, such potential delay times being comparable to the actual wavelength of the signal; and the size of the actual components in the transmitter approaching the wavelength of the signal.